

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. SYSTEM EVOLUTION & BASELINE :

The current MVP (Minimum Viable Product) successfully operates as a highly responsive, AI-driven fitness planner utilizing local SQLite storage and FastAPI routing. To evolve this academic prototype into a robust, commercial health-tech platform, several scaling strategies have been mapped out.

3. STRATEGIES FOR COMMERCIALIZATION & SCALING

- **Database Migration:** Upgrading the data persistence layer from local SQLite to a cloud-hosted PostgreSQL database. Because the application utilizes SQLAlchemy ORM, this transition requires minimal code refactoring.
- **User Authentication & Profiles:** Implementing a secure login system (OAuth 2.0 or JWT). This will allow users to log in, track their historical workout plans, and update their weight over time to receive dynamically adjusting AI routines.
- **Wearable Technology Integration:** Expanding the input parameters by integrating APIs from fitness wearables (e.g., Apple Health, Google Fit). This would allow the Gemini model to analyze real-time heart rate and sleep data to generate even more precise daily workout adjustments.
- **Advanced Load Balancing:** Deploying the FastAPI application using Docker containers orchestrated by Kubernetes to seamlessly handle thousands of concurrent API requests without triggering rate limits.